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| GPS DENIAL, ELECTRONIC JAMMING & DEAD RECKONING EXECUTION DIRECTIVE |
| FLEET: ARDUPILOT EKF3 UNIVERSAL                    AVIONICS OPERATIONAL SOP |
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PART I: AVIONICS PARAMETER TECHNICAL REFERENCE

1. GNSS HARDWARE CUTOFF & RF JAMMING SIMULATION

SIM_GPS1_ENABLE

[HARDWARE TOGGLE]

DEFAULT: 1 (ACTIVE) **TARGET:** 0 (SEVERED / DENIED)

LSK 1L COMMAND: `param set SIM_GPS1_ENABLE 0`

Simulates physical antenna severance or receiver power cutoff. SITL instantly sets $fix_type = 0$ (NO_FIX) and $satellites_visible = 0$. EKF3 immediately declares satellite aiding invalid and transitions directly to smooth Dead Reckoning without coordinate jitter.

SIM_GPS1_JAM

[ELECTRONIC WARFARE INJECTION]

DEFAULT: 0 (NORMAL RF) **TARGET:** 1 (RF JAMMED)

LSK 2L COMMAND: `param set SIM_GPS1_JAM 1`

Injects broadband RF interference on GPS L1/L2 carrier frequencies ($C/N_0 < 28$ dB-Hz), inflating pseudorange noise. CAUTION: SITL maintains a degraded 3D fix with jumping coordinates; requires automated gating configuration (Section 2) to eliminate aircraft wandering.

2. INNOVATION GATING & ATTITUDE HORIZON PROTECTION

EK3_GLITCH_RAD

[INNOVATION STEP GATE THRESHOLD]

DEFAULT: 25.0 m **RECOMMENDED:** 15.0 to 20.0 m

MCDU COMMAND: `param set EK3_GLITCH_RAD 15`

Defines maximum allowable horizontal step jump between consecutive GPS updates before EKF3 rejects the measurement as anomalous. Rejects abrupt adversarial spoofing coordinates and extreme jamming spikes.

FS_EKF_THRESH

[FLIGHT SAFETY FAILSAFE RATIO]

DEFAULT: 0.80 **ANTI-JAM TARGET:** 0.60 to 0.65

MCDU COMMAND: `param set FS_EKF_THRESH 0.60`

Normalized innovation variance ratio that triggers autonomous EKF Failsafe. Lowering to 0.60 forces ArduPlane to trip failsafe immediately under jamming noise rather than tolerating noisy jumping fixes.

AHRS_GPS_USE

[ATTITUDE FILTER ISOLATION]

DEFAULT: 1 (ENABLED) **HOSTILE RF TARGET:** 0 (ISOLATE AHRS)

MCDU COMMAND: `param set AHRS_GPS_USE 0`

Disconnects GPS ground velocity from attitude centrifugal turn compensation ($a_{\text{cent}} = \omega_z \cdot V_{\text{gps}}$). Prevents erratic GPS velocity spikes from corrupting the artificial horizon and inducing violent wing-rocking.

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+-------------------------------------------------------------------------+
| SENSOR FALLBACK CASCADE & AERODYNAMIC DRAG WIND ESTIMATION |
| SUBSYSTEM: EKF3 INERTIAL CORE                                AVIONICS OPERATIONAL SOP |
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3. DEAD RECKONING & AERODYNAMIC WIND ESTIMATION

EK3_DRAG_BCOEF_X / EK3_DRAG_BCOEF_Y

[DRAG BALLISTICS]

DEFAULT: 0.0 (DISABLED) **AIRFRAME TARGET:** 25.0 to 35.0 (kg/m^2)

LSK 4L COMMANDS: param set EK3_DRAG_BCOEF_X 25 | param set EK3_DRAG_BCOEF_Y 25

Calibrates longitudinal and lateral airframe ballistic drag. In GPS-denied flight, EKF3 combines pitot true airspeed (V_{tas}) with accelerometer drag deceleration to estimate horizontal wind ($\vec{W}$), synthesizing ground track: $\vec{V}_{\text{ground}} = \vec{V}_{\text{air}} + \vec{W}$.

EK3_SRC_OPTIONS

[SENSOR FALLBACK CASCADE BITMASK]

DEFAULT: 0 (MANUAL FALLBACK) **AUTONOMOUS TARGET:** 1 (AUTO FALLBACK)

LSK 5L COMMAND: param set EK3_SRC_OPTIONS 1

Enables automatic sensor cascade. Setting Bit 0 allows EKF3 to drop an unhealthy GPS autonomously and switch to Dead Reckoning or Optical Flow without pilot intervention.

4. OPTICAL FLOW SIMULATION & HOVER VELOCITY FUSION

SIM_FLOW_ENABLE / EK3_FLOW_USE / EK3_SRC1_VELXY

[FLOW FUSION]

DEFAULT: 0 / 1 / 3 **EXECUTION TARGET:** 1 / 1 / 5

MCDU COMMANDS: param set SIM_FLOW_ENABLE 1 | param set EK3_SRC1_VELXY 5

Activates simulated downward optical flow camera and directs EKF3 to ingest pixel displacement rates as primary horizontal velocity (V_x, V_y), decoupling hover navigation from satellites.

5. ENVIRONMENTAL WIND & DRYDEN TURBULENCE INJECTION

SIM_WIND_SPD / SIM_WIND_DIR / SIM_WIND_TURB

[METEOROLOGY]

RANGE: SPD: 0--30 m/s | DIR: 0--359° | TURB: 0.0--1.0

MCDU COMMANDS: param set SIM_WIND_SPD 12 | param set SIM_WIND_DIR 270

Injects steady crosswinds and Dryden atmospheric gusts into SITL to validate crab angle computation and EKF3 wind triangle estimation under realistic turbulence.

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+-----------------------------------------------------------------------+
| TACTICAL FLIGHT SCENARIOS 1 & 2 EXECUTION PROTOCOLS |
| TOTAL GNSS HARDWARE CUTOFF & ELECTRONIC WARFARE JAMMING COUNTER |
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SCENARIO 1: TOTAL GNSS HARDWARE CUTOFF [CLEAN DEAD RECKONING]

OBJECTIVE: Evaluate pure aerodynamic and inertial dead reckoning following complete loss of primary satellite reception (antenna severance or hardware failure). Verify aircraft tracks waypoints without wander.

PREREQUISITES: Airspeed calibrated (± 1.0 m/s); cruise in AUTO or GUIDED at 50--100 m AGL; drag coefficients set: EK3_DRAG_BCOEF_X/Y = 25.

LSK 1L TRIGGER COMMAND:

```
param set SIM_GPS1_ENABLE 0
```

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- MAVLink GPS_RAW_INT: fix_type drops immediately to 0 (NO_FIX), $satellites_visible = 0$.
- EKF3 detects instant satellite loss; cleanly decouples horizontal GPS fusion.
- Aircraft enters **Wind Dead Reckoning**: smoothly banks along waypoint legs, holds track via airspeed and compass heading. Zero weaving or wing-rocking occurs.

RECOVERY COMMAND: `param set SIM_GPS1_ENABLE 1` $\rightarrow$ GPS lock reacquired ($fix_type = 3$); fused in 3--5 s.

SCENARIO 2: EW RF JAMMING COUNTERMEASURE [ANTI-HUNT PROTOCOL]

OBJECTIVE: Counter hostile broadband RF noise injection. Prevent aircraft from chasing jumping coordinates (the ''here-and-there'' hunting phenomenon) by forcing automatic innovation rejection and prompt Dead Reckoning fallback.

ROOT CAUSE OF WANDERING: During RF jamming, receiver does *not* declare $fix_type = 0$ immediately. It outputs a degraded 3D fix with jumping coordinates. L1 navigation controller turns toward each false jump, and AHRS_GPS_USE 1 tilts horizon with false ground velocities.

STEP 1 --- EXECUTE ANTI-JAM DEFENSIVE PRE-CONFIGURATION (LSK 5L):

COMMAND	OPERATIONAL FUNCTION
`param set EK3_SRC_OPTIONS 1`	Enables EKF3 automatic fallback away from failing GPS.
`param set FS_EKF_THRESH 0.60`	Tightens failsafe threshold to trip instantly under RF noise.
`param set EK3_GLITCH_RAD 15`	Tightens innovation gate from 25 m to 15 m.
`param set AHRS_GPS_USE 0`	Severs GPS velocity from attitude filter to prevent wing-rocking.

STEP 2 --- INJECT RF JAMMING NOISE (LSK 2L TRIGGER):

```
param set SIM_GPS1_JAM 1
```

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- SITL injects carrier noise: receiver HDOP inflates rapidly (> 4.5) and position variance surges.
- With FS_EKF_THRESH 0.60, EKF failsafe trips within 4--6 seconds.
- With EK3_SRC_OPTIONS 1, EKF3 severs the noisy GPS and cascades into Dead Reckoning.
- With AHRS_GPS_USE 0, wings remain completely level; erratic weaving is eliminated.

RECOVERY SEQUENCE: `param set SIM_GPS1_JAM 0` $\rightarrow$ `param set AHRS_GPS_USE 1`

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| TACTICAL FLIGHT SCENARIOS 3 & 4 EXECUTION PROTOCOLS |
| SPOOFING GATING DEFENSE & OPTICAL FLOW HOVER FALLBACK |
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SCENARIO 3: ADVERSARIAL GPS SPOOFING STEP-JUMP DEFENSE

OBJECTIVE: Validate EKF3 innovation gating defense against hostile terrestrial spoofers attempting to hijack the aircraft trajectory via false pseudoranges or sudden coordinate offset steps.

STEP 1 --- GATING DEFENSIVE PRE-CONFIGURATION: `param set EK3_GLITCH_RAD 15` | `param set AHRS_GPS_USE 0`

STEP 2 --- INJECT COORDINATE STEP OFFSET (LSK 3L TRIGGER):

`param set SIM_GPS1_GLT_X 60` | `param set SIM_GPS1_GLT_Y 60`

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- Spoofed coordinate jump (≈ 85 m) exceeds the EK3_GLITCH_RAD 15 m gate.
- EKF3 flags GPS_Glitch: pos step in MAVLink STATUSTEXT; false vector is rejected.
- Aircraft continues holding planned track on internal dead reckoning, completely ignoring the spoofer's pull.

RECOVERY COMMANDS: `param set SIM_GPS1_GLT_X 0` | `param set SIM_GPS1_GLT_Y 0` | `param set AHRS_GPS_USE 1`

SCENARIO 4: VTOL / LOW-ALTITUDE HOVER OPTICAL FLOW FALLBACK

OBJECTIVE: Ensure stationary hover and station-keeping capability for VTOL platforms or quadplanes when GPS is denied in zero-airspeed hover flight.

STEP 1 --- SENSOR PRE-CONFIGURATION: `param set SIM_FLOW_ENABLE 1` | `param set EK3_FLOW_USE 1` | `param set EK3_SRC1_VELXY 5`

STEP 2 --- FLIGHT REGIME ESTABLISHMENT: Establish aircraft in steady hover (QLOITER or QHOVER mode) at 5--15 m AGL over textured ground.

STEP 3 --- SEVER GPS IN HOVER (LSK 1L TRIGGER):

`param set SIM_GPS1_ENABLE 0`

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- MAVLink OPTICAL_FLOW packets stream angular ground pixel displacement rates (rad/s).
- EKF3 seamlessly substitutes satellite horizontal velocity with integrated optical flow ground velocity.
- Aircraft holds stationary hover coordinates without drift, even with zero forward airspeed over the pitot tube.

RECOVERY COMMANDS: `param set SIM_GPS1_ENABLE 1` | `param set EK3_SRC1_VELXY 3`

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| MASTER FLIGHT TEST MATRIX & COCKPIT QUICK-KEY REFERENCE |
| OPERATIONAL FLIGHT TEST CHECKLIST                                     PAGE: 5 OF 5 |
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SCENARIO 5: HIGH-CROSSWIND TURBULENCE DEAD RECKONING

OBJECTIVE: Validate EKF3 wind triangle estimation ($\vec{V}_{\text{ground}} = \vec{V}_{\text{tas}} + \vec{W}_{\text{wind}}$) and crab-angle compensation under strong crosswind gusts and Dryden turbulence during GPS denial.

STEP 1 --- INJECT CROSSWIND METEOROLOGY: `param set SIM_WIND_SPD 12` | `param set SIM_WIND_DIR 270` | `param set SIM_WIND_TURB 0.25`

STEP 2 --- SEVER GPS UNDER CROSSWIND: `param set SIM_GPS1_ENABLE 0` $\rightarrow$ Aircraft computes crab angle ($\beta \approx 35°$), holding ground track true.

RECOVERY: `param set SIM_GPS1_ENABLE 1` | `param set SIM_WIND_SPD 0`

PART III: MASTER FLIGHT TEST MATRIX

SCENARIO	PRE-SETUP	TRIGGER	KEY TELEMETRY	EVALUATION
1. CLEAN LOSS	Airspeed valid, BCOEF = 25	`param set SIM_GPS1_ENABLE 0`	GPS.fix = 0 GPS.sats = 0	Smooth transition to Dead Reckoning; zero wandering or wing-rock.
2. JAM COUNTER	EK3_SRC_OPT 1 FS_EKF 0.60 AHRS 0	`param set SIM_GPS1_JAM 1`	HDOP > 4.5 FS_EKF in < 6 s	Failsafe trips promptly; drops degraded GPS; zero coordinate hunting.
3. SPOOF DEFENSE	GLITCH_RAD 15 AHRS_GPS 0	`param set SIM_GPS1_GLT_X 60`	GPS_Glitch in STATUSTEXT	Measurement rejected at gate; internal dead reckoning maintained.
4. FLOW HOVER	FLOW_ENABLE 1 SRC1_VELXY 5	`param set SIM_GPS1_ENABLE 0`	OPTICAL_FLOW rate > 20 Hz	Zero positional drift in low-altitude hover; velocity aided by camera.
5. HIGH-WIND DR	WIND_SPD 12 WIND_DIR 270	`param set SIM_GPS1_ENABLE 0`	Crab angle established; track true	Correct crosswind crab computed from drag model; path held in gusts.

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COCKPIT MCDU / LSK QUICK-KEY MAP REFERENCE:
< LSK 1L: Clean GPS Cutoff (SIM_GPS1_ENABLE 0)                LSK 1R >:  EXEC STAGED COMMAND
< LSK 2L: RF Jamming Noise (SIM_GPS1_JAM 1)                   LSK 2R >:  RE-ESTABLISH 3D FIX
< LSK 3L: Spoofing Offset (SIM_GPS1_GLT_X/Y)                 LSK 3R >:  RESET GLITCH OFFSETS
< LSK 4L: Wind Dead Reckoning (BCOEF_X/Y 25)                     LSK 4R >:  RESTORE SENSORS
< LSK 5L: Anti-Jam Config (SRC 1, FS 0.60, AHRS 0)             LSK 5R >:  FLIGHT LOG EXPORT
< LSK 6L: OPEN THIS SOP MANUAL                                 LSK 6R >:  RECOVER ALL NOMINAL
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